Gemma 3 vs. Gemma 4

Key Differences

Feature	Gemma 3 (Released March 12, 2025)	Gemma 4 (Released April 2, 2026)
Licensing	Gemma License (Permissive, but with usage restrictions)	**Apache 2.0** (Fully open-source, unrestricted commercial and enterprise use)
Model Sizes	1B, 4B, 12B, 27B	**E2B, E4B** (Edge models), **26B A4B** (Mixture-of-Experts), and **31B** (Dense)
Architecture	Standard Dense Transformer	**Mixture-of-Experts (MoE)** in the 26B variant; introduces Per-Layer Embeddings (PLE) and Shared KV Caching for memory efficiency.
Context Window	131,072 tokens (128K)	128K (Small models) up to **256K** (26B & 31B); utilizes Dual RoPE (Standard for local, Proportional for global context).
Supported Modalities	Text and Image (1B was text-only)	Text, Image, Video, and **Native Audio** (built-in Conformer architecture for direct speech processing).
Vision Processing	Fixed aspect ratio (squashed/resized images)	**Variable aspect ratio** mapping with a configurable soft token budget and 2D spatial RoPE (maintains native image proportions).
Agentic Capabilities	Standard instruction tuning and function calling	**Native System Prompts**, explicit reasoning parser support, and advanced autonomous tool-use capabilities.
Hardware Targeting	Consumer GPUs to Datacenters	Hyper-optimized for true edge deployment (smartphones, Raspberry Pi) scaling up to multi-GPU enterprise environments.

Test 1:

Quantum Photonics (Explicit Reasoning & Math)

Prompt:

- A Mach-Zehnder interferometer operates at $\lambda = 1550$ nm. Introduce a phase shift $\Delta\phi$ in one arm using a material with an electro-optic coefficient $r_{33} = 30$ pm/V and an applied voltage V.
- Using the <|think|> tag, derive the required voltage to achieve a $P = 0.5$ transmission probability (a 50:50 beam splitter effect)
  - given a device length $L = 1$ cm, refractive index $n = 2.2$, and electrode gap $d = 5$ μm.

Reference Answer Expectation:

- The model must use the reasoning tag to derive the phase shift equation $\Delta\phi = \frac{\pi}{\lambda} n^3 r_{33} \frac{V}{d} L$. It should equate $P = \cos^2(\Delta\phi/2) = 0.5$, solve for $\Delta\phi = \pi/2$, and then isolate V.
- The final computed voltage should be approximately 1.15 V.
- (or, exactly 1.21V)

Comment:
Test passed.

Test 2:

Lens Design & Standards (Large Context Retrieval)

How to check:

1. Upload a massive text file containing the complete ISO 10110 optical drawing standard documentation (testing the 128K/256K context limit).
2. Prompt: Extract the specific notation requirements for surface form tolerances (Part 5) and surface imperfection tolerances (Part 7). Map these constraints to a theoretical AR/VR lens element with a clear aperture of 20 mm.

Reference Answer Expectation:

- The model must not lose details from the beginning of the document (the "lost in the middle" phenomenon).
- It should accurately output standard formats like $3/A(B/C)$ for form tolerances and $5/N \times A$ for imperfections, correctly applying them to the 20 mm aperture specification without hallucinating arbitrary limits.

Comment:
Test passed.

Test 3:

Thermal Photonics (Autonomous Code Generation)

Prompt:

- Generate a standalone Python simulation that iteratively optimizes the thickness of an anti-reflective coating to maximize thermal emissivity at a target wavelength of 10 μm.
- The simulation must save a parameter file, organize outputs by iteration, output plots as images, and generate a final text summary in a timestamped root directory.

Reference Answer

Expectation

- The output must be a single, substantial python artifact. It must utilize thin-film interference equations or the transfer matrix method. It must strictly adhere to the file-system constraints without requiring manual directory creation.

Comment:
OK, but needed to modified/corrected.

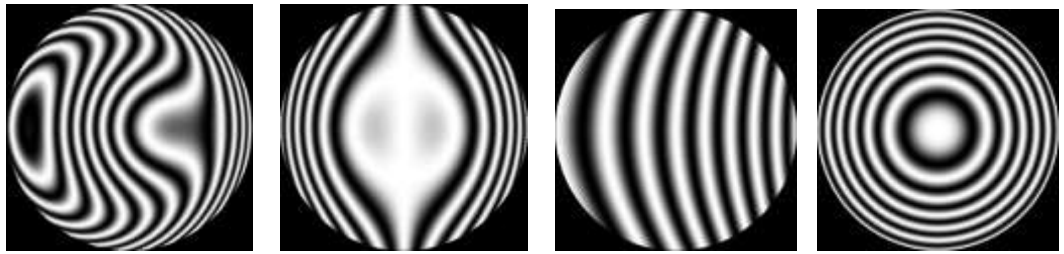
Test Series 4:

The Interferogram (Wavefront Diagnosis)

How to check:

1. *Upload a fringe pattern.*
2. Prompt:
   - Evaluate this interferometric fringe pattern.
   - Identify the primary Zernike polynomial components present (e.g., defocus, spherical, coma).
   - Use the <|think|> reasoning scratchpad to explain your deduction based on fringe curvature and symmetry."

fringe pattern



```
=== Optical Diagram Synthesis Summary ===
Total Diagrams Generated: 4

Upload these generated diagrams to the LLM and prompt it to identify the injected aberration.
-----------------------------------------------
Iter 000 | Aberration: spherical | Magnitude: 1.5 waves | Tilt: 5.0
Iter 001 | Aberration: coma_x | Magnitude: 2.0 waves | Tilt: 3.0
Iter 002 | Aberration: astigmatism_x | Magnitude: 1.0 waves | Tilt: 4.0
Iter 003 | Aberration: defocus | Magnitude: 3.0 waves | Tilt: 0.0
```

Reference Answer Expectation:

- The response must move beyond "curly lines" and correctly correlate specific geometric distortions (e.g., 'S'-shaped fringes) to specific aberrations (coma).

Comment: Test passed.